

Course Code	21EIE602T	Course Name	EMBEDDED CONTROL SYSTEMS	Course Category	E	PROFESSIONAL ELECTIVE	L	T	P	C
							3	0	0	3

Pre-requisite Courses	Nil	Co- requisite Courses	Nil	Progressive Courses	Nil
Course Offering Department	Electronics and Instrumentation Engineering	Data Book / Codes / Standards	Nil		

Course Learning Rationale (CLR):	The purpose of learning this course is to:
CLR-1:	know the significance of embedded controllers in Embedded Systems design
CLR-2:	impart knowledge on human Machine Interfaces and their design constraints
CLR-3:	introduce the significance of PetriNets and Statecharts in Embedded System Modeling

Course Outcomes (CO):	At the end of this course, learners will be able to:	Programme Outcomes (PO)		
		1	2	3
CO-1:	understand the significance of embedded controllers in Embedded Systems design	-	2	-
CO-2:	analyze the importance of HMI in an embedded control system	-	-	3
CO-3:	apply the knowledge of PetriNets and Statecharts in Embedded System Modeling	3	-	3

Module-1 - Embedded Controller Design	15 Hour
Introduction to Embedded systems - Overview of cyber-physical systems - Real Time Embedded Control System - Real time scheduling - I/O Management - Embedded Operating Systems- Networking Protocols - Systems on chip - Memory Subsystem - Bus Structure - Interfacing Protocol - Peripheral Interfacing - Power Management - Embedded System Software Programming Optimization - Concurrent Programming	
Module-2 - Dynamics and Debugging	15 Hour
Continuous dynamics- Discrete dynamics - State machines - Security - Reliability testing - Heuristic evaluation- Interface requirements - Implementation challenges - Processes and real time operating systems - High level simulation - Low level simulation – Onboard debugging - Task level debugging - Emulation - Examples of system design	
Module-3 – Embedded System Modeling	15 Hour
State charts - Modeling Hierarchy - Specification Description Language (SDL) - Petri Nets - Embedded systems modeling with Petri Nets - Unified Modeling Language (UML) - Activity diagram - Class diagram - Component diagram - Use-case diagram - Sequence diagram - UML specification examples - Peripheral Interfaces used in Embedded Systems	

Learning Resources	1. Steve heath, Embedded System Design, 3rd Edition, Packt Publishing, 2013 2. Josheph Yiu, The definitive Guide to ARM Cortex M3 and M4 Processors, 3rd Edition, Elsevier, 2013 3. Shanthanu Chattopadhyay, Embedded System Design, 1stEdition, PHI learning, 2013 4. Prof. Santanu Chaudhary, Introduction to Embedded Computing, NPTEL Course Material, Department of Electrical Engineering, Indian Institute of Technology Delhi, https://nptel.ac.in/courses/108102045 5. Lawrence J. Henschen, Julia C. Lee, "Embedded System Design Methodologies and Issues", Elsevier Science, 2023
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Learning Assessment							
	Bloom's Level of Thinking	Continuous Learning Assessment (CLA)				Summative Final Examination (40% weightage)	
		Formative CLA-1 Average of unit test (50%)		Life-Long Learning CLA-2 (10%)			
		Theory	Practice	Theory	Practice	Theory	Practice
Level 1	Remember	20%	-	20%	-	20%	-
Level 2	Understand	20%	-	20%	-	20%	-
Level 3	Apply	30%	-	30%	-	30%	-
Level 4	Analyze	30%	-	30%	-	30%	-
Level 5	Evaluate	-	-	-	-	-	-
Level 6	Create	-	-	-	-	-	-
	Total	100 %		100 %		100 %	

Course Designers		
Experts from Industry	Experts from Higher Technical Institutions	Internal Experts
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